

ALBERTO LAZO

GIS Developer | AgriTech Specialist | Mobile Geospatial Solutions

AlbertoLazoB@gmail.com | 305-607-5432 | Miami, FL

Green Card Holder | English & Spanish (Native)

LinkedIn: linkedin.com/in/alberto-lazo | GitHub: github.com/betolix

PROFILE

GIS Developer and AgriTech specialist with 10+ years building geospatial solutions for agricultural insurance and land surveying. Expert in mobile GIS applications, satellite imagery analysis, and crop identification using spectral signatures. Proven track record delivering production systems for field data collection, parcel measurement, and agricultural risk assessment. Combines deep GIS expertise with strong software development skills.

TECHNICAL SKILLS

GIS & Geospatial: QGIS, Google Maps API, GeoJSON, GPS/Location Services, satellite imagery analysis, spectral signatures

Mobile Development: Android (Kotlin, Java), Jetpack Compose, Room, Retrofit, offline-capable apps

Data Processing: Python, imagery analysis, crop classification algorithms

Backend: Spring Boot, Node.js, PostgreSQL, REST APIs

PROFESSIONAL EXPERIENCE

Solutions Architect / Mobile GIS Lead | *Pacífico Seguros*

Jan 2020 - Nov 2022 | Remote

Designed and delivered mobile geospatial solutions for agricultural insurance programs in Peru.

- Built native Android applications for field surveying and crop inspection using Java, Google Maps, and GeoJSON
- Integrated GPS sensors, satellite imagery, and statistical sector mapping for accurate land measurement
- Developed offline-capable data collection with automatic sync to backend systems
- Enabled insurers to assess parcel size, crop type, and growth stage for catastrophic agricultural insurance
- Deployed to production with a team of 5 field inspectors covering agricultural regions

Technologies: Android (Java), Google Maps API, GeoJSON, Room, Retrofit, GPS/Location Services

Software Development Lead - GIS Focus | *Pacífico Seguros*

Jun 2012 - Dec 2019 | Remote

Led development of geospatial and agricultural technology solutions for insurance operations.

- Implemented satellite imagery analysis using QGIS and Python to determine crop types through spectral signatures
- Led team of 2 developers building mobile crop inspection applications
- Processed drone and satellite imagery for Agricultural Catastrophic Insurance Program of Peru
- Developed algorithms for crop identification and growth stage assessment

Technologies: Python, QGIS, satellite imagery, spectral analysis, Android (Java)

AI Workflow Architect / Tech Lead | *Pacífico Seguros*

Nov 2022 - Feb 2025 | Remote

- Applied image classification expertise to AI-based property claims assessment
- Designed business rules for automated image processing systems

Technologies: AI/ML workflows, image classification, business rules engines

Business Systems Consultant | *Datapro Miami*

Mar 2025 - Aug 2025 | Miami, FL

- Led Jira Service Management implementation and PowerBI integration

Technologies: Jira Service Management, PowerBI

GIS/AGRITECH PROJECTS

Land Survey Platform - Production Mobile GIS Application

- Mobile application for agricultural land surveying with offline capability
- Google Maps integration with GeoJSON-defined statistical sectors
- GPS-based parcel measurement and crop area calculation
- Automatic data synchronization with backend systems

GitHub: github.com/betolix/LandSurveyPlatform

AgroScout - Agricultural Data Collection Platform (In Development)

- Comprehensive GIS application for parcel and subparcel management
- Vineyard management including phytonutritional applications and phenological stages
- Multi-platform: Android (Kotlin), web dashboard, offline capability

GitHub: github.com/betolix/AgroScout

DOMAIN EXPERTISE

- Agricultural Insurance: Crop inspection, risk assessment, catastrophic coverage programs
- Remote Sensing: Satellite imagery analysis, spectral signatures, crop classification
- Land Surveying: Parcel measurement, GPS integration, field data collection
- Phenology: Crop growth stages, agricultural calendars, harvest timing

EDUCATION

Full Stack Software Developer | 4Geeks Academy | 2022 - 2023

Bachelor in Informatic Engineering | Universidad Católica San Pablo | 2000 - 2005